# Akoizo Original Practice - Form 5 - Answer key

### Division C | Electric Vehicle | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

#### 1. (2 points) Which pusher feature is NOT allowed?

Unit: The bottle pusher: catching, pushing, and the narrow-opening bonus.
Worked-case data: Compare pushers with openings of 9.0 cm and 15.0 cm. The 9.0 cm version catches the bottle in 8 of 10 tests; the 15.0 cm version in 10 of 10. A miss costs about 420 points (400 cm bottle penalty plus no bonus).
Separate investigation observations:
Lateral error at bottle: Up to ±2.0 cm in 20 runs.
Pusher A: 8.0 cm opening, no funnel: 14 of 20 captures.
Pusher B: 11.0 cm opening with V entry: 20 of 20 captures.

A. Hinged arms that close around the bottle

B. A rigid V-shaped funnel

C. Rounded front edges

D. A curved pocket

**Correct: A**

#### 2. (2 points) Why are quadrature encoders useful in this event?

Unit: Drivetrain and distance sensing: motors, gears, and encoders.
Worked-case data: A 6.5 cm wheel drives a 100-count-per-revolution encoder through a 3:1 motor-to-wheel gear ratio, with the encoder on the motor shaft. How many counts for 8.60 m?
Separate investigation observations:
Forward test: Commanded 500 cm, measured 503.5 cm.
Reverse test: Commanded 150 cm back, measured 148.9 cm.
Tires: Rubber O-rings, slightly compressed under load.

A. They detect direction, which matters when the vehicle reverses

B. They measure battery voltage

C. They replace the pencil trigger

D. They measure the bottle distance

**Correct: A**

**3. (2 points) The bottle never gets past the Target Point. What Bottle Distance is used?**

A. 400.0 cm

B. 0 cm

C. The measured distance

D. 2500 cm

**Correct: A**

**4. (2 points) A team uses a kit with only one documented modification. What happens?**

A. No penalty

B. A 300-point construction violation

C. A 50-point non-modification penalty on its runs

D. Participation points only

**Correct: C**

**5. (4 points) Why is closed-loop speed control better than fixed power for timing?**

answer: Fixed power gives a speed that changes with battery voltage and load; feedback measures speed and corrects it, keeping run time consistent. (4 points).

**6. (4 points) Why should the program take the posted values as inputs rather than hard-code them?**

answer: The Target Distance, Bottle Line Distance, and Target Time are announced at each tournament, so the program must compute the path for any legal values. (4 points).

**7. (4 points) Why does the supervisor pause event time before each run?**

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

answer: To check the vehicle in its ready-to-run configuration for violations without using the team's time. (4 points).

**8. (4 points) Why push low on the bottle?**

Unit: The bottle pusher: catching, pushing, and the narrow-opening bonus.
Worked-case data: Compare pushers with openings of 9.0 cm and 15.0 cm. The 9.0 cm version catches the bottle in 8 of 10 tests; the 15.0 cm version in 10 of 10. A miss costs about 420 points (400 cm bottle penalty plus no bonus).
Separate investigation observations:
Lateral error at bottle: Up to ±2.0 cm in 20 runs.
Pusher A: 8.0 cm opening, no funnel: 14 of 20 captures.
Pusher B: 11.0 cm opening with V entry: 20 of 20 captures.

answer: Pushing near the base reduces the tipping moment, keeping the tall bottle upright. (4 points).

**9. (4 points) Why choose a gear ratio with speed margin?**

Unit: Drivetrain and distance sensing: motors, gears, and encoders.
Worked-case data: A 6.5 cm wheel drives a 100-count-per-revolution encoder through a 3:1 motor-to-wheel gear ratio, with the encoder on the motor shaft. How many counts for 8.60 m?
Separate investigation observations:
Forward test: Commanded 500 cm, measured 503.5 cm.
Reverse test: Commanded 150 cm back, measured 148.9 cm.
Tires: Rubber O-rings, slightly compressed under load.

answer: The vehicle must cover the full path within the shortest possible Target Time, even with low batteries and pushing a bottle. (4 points).

**10. (6 points) Solve this case and explain your reasoning: Target Time 14 s. Pusher opening 16.0 cm. The vehicle stops 22.5 cm from the Target Point in 16.37 s; the bottle ends 12.1 cm beyond the Bottle Line. Find the Run Score.**

reasoning: Distance Score = 22.5 x 2.0 + 12.1 x 1.0 = 57.10.
Time Score = |14 - 16.37| x 0.5 = 1.19; Bottle Bonus = -20.
Pusher Bonus = (16.0 - 35.0) x 1.5 = -28.50; total = 100 + 57.10 + 1.19 - 20 - 28.50. (3 points).

conclusion: Run Score = 109.79, matching the rules' scoring example. (3 points).

**11. (6 points) Solve this case and explain your reasoning: A team's vehicle uses six NiMH AA cells, a motor driver chip, a microcontroller, a 1 F supercapacitor to boost acceleration, and a laser pointer for aiming. What must change?**

reasoning: Six labeled NiMH AAs, a motor driver, and a microcontroller are allowed.
The supercapacitor is an energy storage device that can exceed battery power: remove it.
The laser pointer is prohibited: replace it with a non-laser sighting device. (3 points).

conclusion: Remove the supercapacitor and the laser; the rest is legal. (3 points).

**12. (2 points) What does the comparison show?**

Supplied synthetic observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

A. Feedback control makes time and stopping independent of battery state

B. Fresh batteries are always slower

C. Fixed power is more accurate

**Correct: A**

**13. (2 points) Define “Target Distance” in the context of Electric Vehicle.**

answer: Start Point to Target Point, 5.00-9.00 m. (2 points).

**14. (2 points) Define “Impound” in the context of Electric Vehicle.**

answer: Turning in the vehicle and materials before the first time slot. (2 points).

**15. (2 points) Define “Widest opening” in the context of Electric Vehicle.**

answer: Measured mouth of the pusher, used for the bonus. (2 points).

**16. (2 points) Define “Quadrature” in the context of Electric Vehicle.**

answer: Two offset encoder signals that reveal direction. (2 points).

**17. (2 points) Define “Bottle Bonus” in the context of Electric Vehicle.**

answer: -20 points when the bottle ends fully beyond the Bottle Line. (2 points).

**18. (2 points) Define “Single-source program” in the context of Electric Vehicle.**

answer: Code copied whole from one source; prohibited. (2 points).